



A predictor for square multivariable dead-time systems with multiple delays based on the Kalman filter

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ABSTRACT

Dead-time is a phenomena that is present in many industrial processes and it presents a challenge for feedback control, especially for multivariable processes. To attenuate the dead-time effects, a common method is the use of predictor structures, which use input/output information of the process to predict the output (or states) of the system after the dead-time. In this paper, the Modified Kalman Predictor (MKP) is proposed, which is a novel predictor for linear multivariable square systems with multiple dead-time (or delays) based on the Kalman Filter that has disturbance estimation and can cope with systems of any order or dynamics, including unstable ones. It uses a specific state-space representation of the process which makes its implementation more straight-forward when compared to other methods. The MKP affects the disturbance rejection but not the closed-loop stability in the nominal case, and it can help to improve closed-loop robustness in the uncertain case. The impacts of the MKP tuning in the closed-loop response considering disturbance rejection and robustness are analyzed using standard frequency domain tools. To illustrate the benefits of the MKP, two examples are used that highlight the tuning guidelines for disturbance rejection and robustness improvements.

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1. Introduction

Dead time (or delay) is an important phenomena that is present in many processes and can be associated with information, energy or mass transportation, and it can also be used as a tool to approximate high-order systems with low-order ones [1]. The delay usually complicates the feedback controller design. The reasons for this are intuitive [2]: (a) when the input of the process changes, its effect will be felt in the output only after the dead time; (b) disturbances are not felt immediately; (c) because of (a) and (b), the controller tries to correct something that happened in the past. Or, interpreting in a mathematical way, in linear time-invariant systems with delays, the closed-loop characteristic equation is not polynomial anymore in the continuous-time case and, in the discrete-time case, the order of the system increases proportionally to the size of the delay. In both cases, the presence of delay makes the system harder to control with traditional techniques [1]. This issue becomes even more problematic in Multiple-Input-Multiple-Output (MIMO) systems due to the increased number of variables, multiple delays and coupling effects.

There are many methods that help to deal with the presence of dead-time in the system. Nonetheless, as shown in [3], they can

all be analyzed through an observer–predictor (O&P) structure (Fig. 1). The core idea is to use a model of the process and the information from the inputs and outputs to predict the states after the dead time. Then, this signal is used to feedback the controller and close the loop. Hence, through the perspective of the feedback controller (also called primary controller) the effects of the dead time are considerably reduced, simplifying the controller design. In the particular case of Single-Input-Single-Output (SISO) systems, the primary controller can be designed considering only the dead-time-free model in the nominal case, since the delay does not appear in the closed-loop characteristic equation [1]. In the general MIMO case, the dead time usually cannot be ignored completely with most methods, but the use of an O&P is still advantageous since it reduces the delay effects in the nominal case.

The most famous observer–predictor method is arguably the Smith Predictor (SP) which was first proposed for SISO systems with transfer function models [4]. Many authors have proposed different solutions for some of the limitations of the SP during the last decades, such as slow disturbance rejection and application to stable systems only. One of these works is the proposal in [5], which improves the disturbance rejection capabilities of the SP and can also be used to control unstable plants. An interesting review of the works done until 1996 can be found in [2], and a more recent review that focuses also on the discrete domain solutions is presented in [1,6]. One of these discrete solutions is the Filtered

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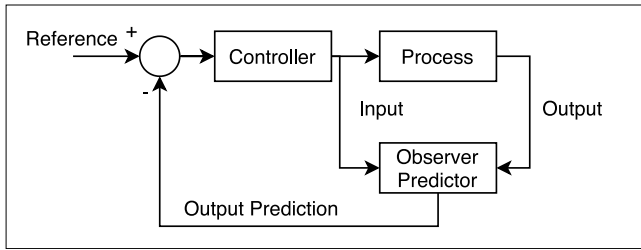


Fig. 1. Observer–predictor structure. Figure based on [2].

Smith Predictor (FSP), which includes a filter in the SP structure that adds a degree of freedom that can be tuned to cope with unstable plants, increase the closed-loop robustness or improves the disturbance rejection capabilities of the SP. However, the FSP application to higher order systems can be involved, and some works try to work around this with particular tuning guidelines or simplified assumptions to achieve better disturbance rejection or robustness [7,8].

Other methods include the Generalized Predictor (GP) [9,10] and the Simplified Generalized Predictor (SGP) [11]. The SGP uses a Finite Impulse Response (FIR) filter and a modified system model to predict the system response, and intrinsically deals with unstable and integrative processes, while also having more straight-forward tuning rules than the FSP. Note that FIR elements are an intrinsic part of many previous proposals, such as of works of [5,12–15]. And then there are ones which use state-space observers such as the steady-state Kalman filter [16–18] or the Luenberger observer, which is specially interesting because their design is less complex [19].

Also, it is worth noting that some advanced control algorithms, such as Model Predictive Control (MPC), implicitly define an O&P structure, as shown in [1] for the specific case of the Generalized Predictive Control (GPC) and in [20] for the Dynamic Matrix Control (DMC). The GPC defines an optimal predictor which makes the stability margins of the controller dependent on the nominal dead time which is related to the implicit disturbance model and observer. Meanwhile, the DMC implicitly defines an SP which explain its slow disturbance rejection for dead-time systems.

The application of predictors to MIMO processes is more difficult than in the SISO case since the coupling effects and multiple delays make it harder to apply some of the techniques developed for the SISO case, and the tuning guidelines become more involved. A unified approach (controller and predictor tuning) based on the FSP is provided in [21] for MIMO square processes, and for MIMO non-square processes in [22]. But, as the number of variables and order of the transfer functions increase, the tuning becomes more difficult. Something similar is done in [23], the controller is simplified to a gain and the MIMO FSP is tuned to achieve the desired closed-loop properties, but the conditions that the MIMO FSP must satisfy are more complex. In [24], the authors suggest the addition of a decoupler to the control structure, but to apply the proposed strategy the system has to satisfy certain conditions.

Some approaches use a state-space representation of the process to obtain the predictor, which is naturally MIMO. However, the prediction is usually obtained through the integration of the system model, which becomes complicated or non-causal in the presence of multiple delays, limiting the application of these techniques considerably. This happens with the approaches presented in [17,25–27]. A solution is shown in [28] for multiple delays, but the predictor has a more complicated structure, it uses a model that always results in a large number of states, and it does not have degrees of freedom to adjust robustness and disturbance rejection, which are not thoroughly analyzed.

In this work, a modified version of the Kalman Predictor for SISO plants presented in [17] is proposed for MIMO square processes that has the following advantages: (i) can deal with multiple delays; (ii) it can be applied to processes of any number of inputs/outputs; (iii) the predictor deals with systems of any order or dynamics (e.g. integrative, unstable and non-minimum phase) without particular rules for each case; (iv) it can be used with state-space or transfer functions models; (v) it does not need to integrate the model to obtain the predictions after the delay; (vi) the tuning of the predictor parameters are intuitive and can be used to improve disturbance rejection or robustness; (vii) the predictor is defined in discrete-time, making the implementation straight-forward; (viii) it can be interpreted as a MIMO FSP, allowing the use of well known analysis tools, specially the frequency domain ones. However, it is important to note that, similarly as in other DTC structures, these advantages depend on how the delays are present in the MIMO process model, which will be discussed in the next section.

Furthermore, the properties of the predictor structure are analyzed independently of the design of the primary controller used in the MIMO DTC structure in order to understand how the predictor affects the closed-loop behavior. Different design techniques are used to compute the primary controller in each example, but the same primary controller is always used in such a way that the predictor properties are highlighted.

The remaining of this work is divided in the following way: the MIMO FSP is presented in Section 2; the model used with the proposed predictor is shown in Section 3; the Modified Kalman Predictor is presented in Section 4, and its analysis in Section 5; results and conclusions are discussed in Sections 6 and 7, respectively.

Notation: $\mathbf{x}(k+j|k)$ refers to the prediction of $\mathbf{x}$ at time $k+j$ given information up to time k . Bold small case letters ($\mathbf{x}$) and upper case letters ($\mathbf{A}$) indicate, respectively, vectors and matrices. $\text{diag}(\cdot)$ represents a diagonal matrix where its non-zero elements are the input parameters. $\mathbf{I}_j$ and $\mathbf{0}_j$ are an identity matrix, and a zero matrix, respectively, of dimension j or, if j is omitted, of appropriate dimensions. z and s are the frequency domain variables of the Z-Transform, and of the Laplace Transform, respectively. $\bar{\sigma}(\mathbf{A})$ is the maximum singular value of $\mathbf{A}$ obtained from the Single Value Decomposition (SVD). $\mathcal{Z}\{\cdot\}$ is the Z-transform of a signal or equation.

2. The Filtered Smith Predictor

The Filtered Smith Predictor (FSP) is an important predictor studied in various works [1,21,29,30], and is the basis for comparison in many others [8,11,31,32]. Its properties are well defined mathematically and the other predictors structures usually can be interpreted as an FSP. Hence, all analysis tools available for the FSP can be used with new predictors, making the analysis of their properties more straight-forward.

First, consider a general MIMO process with multiple delays described by its discrete transfer function matrix $\mathbf{P}(z)$ in the following way:

$$\mathbf{P}(z) = \begin{bmatrix} G_{11}(z)z^{-d_{11}} & \cdots & G_{1m}(z)z^{-d_{1m}} \\ \vdots & \ddots & \vdots \\ G_{n1}(z)z^{-d_{n1}} & \cdots & G_{nm}(z)z^{-d_{nm}} \end{bmatrix} \quad (1)$$

where n , and m are the number of outputs and inputs, respectively. $G_{ij}(z)$ is a delay-free transfer function between output i and input j , while d_{ij} is the discrete delay between them. Since only square MIMO processes will be dealt with here, $n = m$. The system (1) can then be described as $\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z)$ where $\mathbf{L}(z)$ is

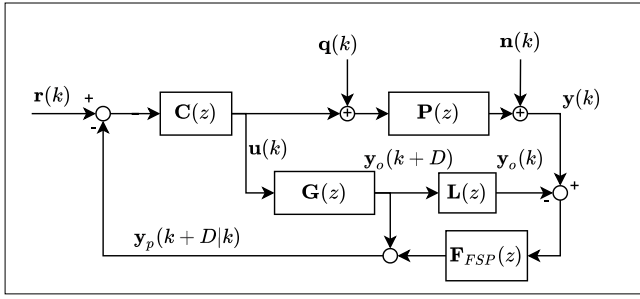


Fig. 2. MIMO Filtered Smith Predictor (FSP) block diagram.

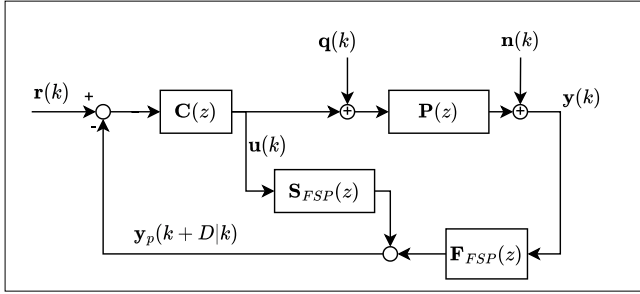


Fig. 3. Implementation structure of the MIMO FSP for unstable and integrative systems.

a diagonal transfer function matrix that represents the minimal output delay of the system, and $G(z)$ is called the fast model:

$$L(z) = \begin{bmatrix} z^{-d_1} & 0 & 0 \\ 0 & \ddots & \vdots \\ 0 & 0 & z^{-d_n} \end{bmatrix} \quad (2)$$

$$G(z) = \begin{bmatrix} G_{11}(z)z^{-(d_{11}-d_1)} & \dots & G_{1n}(z)z^{-(d_{1n}-d_1)} \\ \vdots & \ddots & \vdots \\ G_{n1}(z)z^{-(d_{n1}-d_n)} & \dots & G_{nn}(z)z^{-(d_{nn}-d_n)} \end{bmatrix} \quad (3)$$

where d_i is the minimal delay of output i or the minimum delay of the i th line of $P(z)$. The model output is defined as $y_o(k)$ and the output of the fast model is $y_o(k+D)$, which is the predicted value of the output before the minimal delays: $y_o(k+D) = [y_1(k+d_1) \ y_2(k+d_2) \ \dots \ y_n(k+d_n)]^T$, and D is used to shorten the notation and to indicate that each output has a different delay.

In the MIMO case, the FSP (Fig. 2) uses the fast model of the process $G(z)$ to compute the predicted value of the output before the minimal delays, $y_o(k+D)$. This predicted value is sent via feedback to a primary controller $C(z)$, which has order $n \times n$. However, this prediction must be corrected in some way to take into account disturbances and model mismatch. In the FSP, this is done by calculating the prediction error, i.e. the difference between the real output of the process $y(k)$ and the output model $y_o(k)$, which is then filtered by $F_{FSP}(z)$ (with dimension $n \times n$), and finally added to the nominal prediction. The output of the filter is defined as $y_f(k)$ and the final corrected prediction used in the primary controller is $y_p(k+D|k)$. Also, in Fig. 2, $q(k)$ is an input disturbance, $n(k)$ is an output disturbance, or noise, $r(k)$ is the reference and $u(k)$ is the process input. Note that the corrected output prediction can be written as: $y_p(k+D|k) = [y_{p1}(k+d_1|k) \ \dots \ y_{pn}(k+d_n|k)]^T$.

The main properties of the MIMO FSP are listed below [1].

- The predictor output signal can be represented in the frequency domain as

$$Y_p(z) = (I - F_{FSP}(z)L(z))G(z)U(z) + F_{FSP}(z)Y(z) \quad (4)$$

with $Y_p(z) = \mathcal{Z}\{y_p(k+D|k)\}$.

- The closed-loop matrix transfer functions $H_r(z)$, $H_q(z)$ and $H_n(z)$, respectively associated to the reference, disturbance and noise are:

$$H_r(z) = P(z)C(z)(I + G(z)C(z))^{-1}, \quad (5)$$

$$H_q(z) = (I - P(z)C(z)(I + G(z)C(z))^{-1}F_{FSP}(z))P(z), \quad (6)$$

$$H_n(z) = (I - P(z)C(z)(I + G(z)C(z))^{-1}F_{FSP}(z)), \quad (7)$$

and note that their characteristic equations are not influenced by the minimal output delays $L(z)$ in the nominal case, thus, the tuning of the primary controller $C(z)$ can be done taking into account only the fast model $G(z)$, which can considerably simplify the design procedure. This happens because $G(z)$ will always have a smaller order than $P(z)$, except when the minimal output delays are all zero but, in this case, the advantages of any DTC structure are diminished. Also note that the difference between the delays in $P(z)$ and $G(z)$ depends on the problem. The most favorable condition, i.e. the greater delay reduction in the elements of a row of $G(z)$, is achieved when the differences between the delays of the elements in the correspondent row of $P(z)$ is small. The tuning of $C(z)$ will not be the subject of this article, but tuning guidelines can be found in [1,33,34].

- The reference tracking is not affected by the prediction-error filter $F_{FSP}(z)$ in the nominal case.
- The disturbance rejection can be adjusted by adequately tuning $F_{FSP}(z)$. For example, in the case of input disturbances (6), the open-loop model $P(z)$ appears as a factor of the matrix transfer function, making the disturbance rejection as slow as the open-loop dynamics. This issue can be avoided if the filter $F_{FSP}(z)$ is diagonal and each of its non-null element satisfy the following condition [21]:

$$\left. \frac{d^\lambda}{dz^\lambda} (1 - F_{FSP,i}(z)z^{-d_i}) \right|_{z=z_j} = 0, \quad \lambda = 0, \dots, m_j-1, \quad j = 1, \dots, k \quad (8)$$

where $F_{FSP,i}(z)$ is the i th diagonal element of $F_{FSP}(z)$, z_j is the j th undesirable pole of the i th row of $G(z)$, and m_j is the multiplicity of z_j .

- If the system $P(z)$ has unstable or integrative poles, the MIMO FSP must be implemented using an alternative structure, shown in Fig. 3, to guarantee the internal stability of the predictor, where:

$$S_{FSP}(z) = (I - F_{FSP}(z)L(z))G(z), \quad (9)$$

and $F_{FSP}(z)$ must be tuned to obtain a stable transfer matrix $S(z)$ by using (8) with any unstable or integrative poles of $G(z)$. In this case, the predictor output can be represented in the frequency domain as:

$$Y_p(z) = S_{FSP}(z)U(z) + F_{FSP}(z)Y(z) \quad (10)$$

- Consider an unstructured additive uncertainty $E(z) = W_2(z)\Delta(z)W_1(z)$ such that $P(z) = P_n(z) + E(z)$, where $P_n(z)$ is the nominal model and $\|\Delta(z)\|_\infty \leq 1$. If the following sufficient condition is satisfied, the closed-loop system remains stable for $P(z)$ [1,21,34],

$$\bar{\sigma}(W_1(z)M_0(z)W_2(z)) \leq 1, \quad \forall z = e^{\pm j\omega T_s}, \quad \omega \in [0, \pi/T_s], \quad (11)$$

where $M_0(z) = C(z)(I + G(z)C(z))^{-1}F_{FSP}(z)$ is stable. Hence, the closed-loop system can become more robust if $F_{FSP}(z)$ has low-pass characteristics, and vice-versa. This condition

is sufficient but not necessary for robustness, i.e. the system might be stable even if the condition is not satisfied, hence, it is a conservative condition, but this usually is not an issue since, in practice, we tend to prefer a larger robustness margin. Also, in here, we consider an additive representation of the uncertainties, but there are more ways that the same uncertainty can be represented, as discussed in [34]. Each representation will have a different frequency domain characteristic which will give different robustness conditions for the same modeling error, thus, the user should be cautious about this choice and choose the error structure that has the least magnitude in the frequency domain, otherwise, there is a risk of being too conservative, which will limit, for example, settling time specifications.

3. System model

The predictor that will be proposed requires, as a first step, a specific state-space representation of the process, which can easily be obtained from the nominal transfer function matrix of the process. As described earlier, the process transfer function is given by $\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z)$. In state-space form, $\mathbf{G}(z)$ and $\mathbf{L}(z)$ can be written, respectively, as

$$\mathbf{G}(z) \rightarrow \begin{cases} \mathbf{x}(k+1) &= \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k) \\ \mathbf{y}_o(k+D) &= \mathbf{C}\mathbf{x}(k) \end{cases} \quad (12)$$

$$\mathbf{L}(z) \rightarrow \begin{cases} \mathbf{x}_l(k+1) &= \mathbf{A}_l\mathbf{x}_l(k) + \mathbf{B}_l\mathbf{y}_o(k+D) \\ \mathbf{y}_o(k) &= \mathbf{C}_l\mathbf{x}_l(k) + \mathbf{D}_l\mathbf{y}_o(k+D) \end{cases} \quad (13)$$

where $\mathbf{y}_o, \mathbf{u} \in \mathbb{R}^n$, $\mathbf{x} \in \mathbb{R}^{n_{xG}}$, $\mathbf{x}_l \in \mathbb{R}^{n_{xL}}$, with n_{xL} and n_{xG} being, respectively, the number of states of $\mathbf{L}(z)$ and $\mathbf{G}(z)$ in state-space form. Both systems can be combined to generate a non-minimal state-space model of $\mathbf{P}(z)$ as shown below:

$$\begin{cases} \begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_l(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(k) \\ \mathbf{y}_o(k) = \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix} \end{cases} \quad (14)$$

An important characteristic of this representation is that the output without the minimal output delay can be obtained directly from the states with

$$\mathbf{y}_o(k+D) = \begin{bmatrix} y_1(k+d_1) \\ y_2(k+d_2) \\ \vdots \\ y_n(k+d_n) \end{bmatrix} = \begin{bmatrix} \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \end{bmatrix}, \quad (15)$$

which can be used by the proposed predictor to directly obtain the predictions after the minimum output delays for processes with multiple dead-times. It is also important to emphasize that in a MIMO system we can consider measurement (output), control or internal delays, and each have a different physical meaning. However, they can all be reinterpreted as output delays to use the proposed formulation.

4. Modified Kalman Predictor for MIMO square processes

The proposed Modified Kalman Predictor (MKP) for square MIMO systems with multiple delays uses a Kalman Filter to estimate the states of the system (14). While estimating the states, it is important to consider the effect of unknown disturbances which can represent both model mismatch and external signals. Their estimation is essential to obtain an offset-free prediction of the system states and/or outputs. A generic state-space model for disturbances is given below,

$$\begin{cases} \mathbf{x}_\phi(k+1) &= \mathbf{A}_\phi\mathbf{x}_\phi(k) \\ \mathbf{y}_\phi(k) &= \mathbf{C}_\phi\mathbf{x}_\phi(k) \end{cases} \quad (16)$$

where $n_{x\phi}$ is the number of disturbance states, $\mathbf{x}_\phi \in \mathbb{R}^{n_{x\phi}}$, $\mathbf{A}_\phi \in \mathbb{R}^{n_{x\phi} \times n_{x\phi}}$ represent the disturbance dynamics, for example, a constant or sinusoid, $\mathbf{C}_\phi \in \mathbb{R}^{n_{y\phi} \times n_{x\phi}}$, and $\mathbf{y}_\phi \in \mathbb{R}^{n_{y\phi}}$ is the actual disturbance that will affect the system.

The Kalman filter has an stochastic formulation in which zero-mean additive Gaussian noises are considered in the states ($\mathbf{w} \in \mathbb{R}^{n_w}$) and output ($\mathbf{v} \in \mathbb{R}^n$). They are defined by diagonal covariance matrices $\mathcal{Q}$ and $\mathcal{R}$, respectively, which means that the noises are uncorrelated. Then, the augmented model that considers the disturbance dynamics is obtained by combining (14) and (16), and assuming that $\mathbf{y}_\phi(k)$ is added to the control input $\mathbf{u}(k)$:

$$\begin{cases} \begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_l(k+1) \\ \mathbf{x}_\phi(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B}\mathbf{C}_\phi \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_\phi \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \\ \mathbf{x}_\phi(k) \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \mathbf{u}(k) \\ \quad \quad \quad + \mathbf{H}\mathbf{w}(k) \\ \mathbf{y}_o(k) = \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_l(k) \\ \mathbf{x}_\phi(k) \end{bmatrix} + \mathbf{v}(k) \end{cases} \quad (17)$$

or in compact form,

$$\begin{cases} \bar{\mathbf{x}}(k+1) = \bar{\mathbf{A}}\bar{\mathbf{x}}(k) + \bar{\mathbf{B}}\mathbf{u}(k) + \mathbf{H}\mathbf{w}(k) \\ \mathbf{y}_o(k) = \bar{\mathbf{C}}\bar{\mathbf{x}}(k) + \mathbf{v}(k) \end{cases} \quad (18a)$$

with

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B}\mathbf{C}_\phi \\ \mathbf{B}_l\mathbf{C} & \mathbf{A}_l & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_\phi \end{bmatrix}, \quad \bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix},$$

$$\bar{\mathbf{C}} = \begin{bmatrix} \mathbf{D}_l\mathbf{C} & \mathbf{C}_l & \mathbf{0} \end{bmatrix},$$

and $\mathbf{H} \in \mathbb{R}^{n_H \times n_w}$, with $n_H = n_{xG} + n_{xL} + n_{x\phi}$. Matrix $\mathbf{H}$ selects where the noise $\mathbf{w}$ appears in the state equations. As stated before, for the proposed MKP algorithm, we consider that the disturbances affect the system as if they were inputs, this is done because disturbances due to model mismatch are dependent on control input, thus, they are better represented with this consideration. In this work, the focus will be only on square MIMO process with constant disturbances, thus, $n_\phi = n_{x\phi} = n$ and $\mathbf{C}_\phi = \mathbf{A}_\phi = \mathbf{I}$, but keep in mind that particular problems might be subject to different types of disturbances such as sinusoids or ramps.

When designing a Kalman filter, the standard approach is to define one noise signal per state, i.e. $n_w = n_{xG} + n_{xL} + n_{x\phi}$. However, this procedure makes the tuning procedure more involved because the number of states can be big for MIMO systems. Thus, following the suggestion in [17], only one noise signal will be added per input. In this simplified case, the interest lies only in the system output estimation after the dead time, and not on the state estimation, hence, the noise in the states can be removed. With these considerations, matrix $\mathbf{H}$ has the following structure

$$\mathbf{H} = \begin{bmatrix} \mathbf{0}_{n_{xG}+n_{xL} \times n} & \mathbf{I}_{n \times n}^T \end{bmatrix}^T.$$

With the augmented system, any software tool can be used to compute the steady-state (fixed gain) Kalman filter to estimate the system states. These tools generally return $\mathbf{L}, \mathbf{M} \in \mathbb{R}^{n_{xG}+n_{xL}+n_{x\phi} \times n}$ which are, respectively, the filter gain and the innovation gain with $\mathbf{L} = \bar{\mathbf{A}}\mathbf{M}$. This procedure can be done as long as the pair $(\bar{\mathbf{A}}, \bar{\mathbf{C}})$ is detectable, which guarantees that a marginally stable steady-state Kalman filter exists ([35], Theorem 26).

Then, the steady-state Kalman Filter can be written as

$$\begin{cases} \bar{\mathbf{x}}(k+1|k) = (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}) \bar{\mathbf{x}}(k|k-1) + [\bar{\mathbf{L}} \quad \bar{\mathbf{B}}] \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \end{cases} \quad (19a)$$

$$\begin{cases} \bar{\mathbf{x}}(k|k) = (\mathbf{I} - \bar{\mathbf{M}}\bar{\mathbf{C}}) \bar{\mathbf{x}}(k|k-1) + [\bar{\mathbf{M}} \quad \mathbf{0}] \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \end{cases} \quad (19b)$$

$$\mathbf{y}_p(k|k) = \bar{\mathbf{C}}\bar{\mathbf{x}}(k|k) \quad (19c)$$

$$\mathbf{y}_p(k+D|k) = [\mathbf{C} \quad \mathbf{0} \quad \mathbf{0}] \bar{\mathbf{x}}(k|k) \quad (19d)$$

where $\bar{\mathbf{x}}(k|k)$, $\mathbf{y}_p(k|k)$, and $\mathbf{y}_p(k+D|k)$ are, respectively, the updated estimation of the system states, the estimated system output and the estimated system output without the minimal output delays. Remember that the latter can be obtained simply by adequately selecting part of the augmented estimated states as explained in Section 3. Since the point of the predictor is to obtain the estimated output after the minimal delays, the Kalman Predictor is obtained from (19) after combining some of its terms:

$$\begin{cases} \bar{\mathbf{x}}(k+1|k) = (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}) \bar{\mathbf{x}}(k|k-1) + [\bar{\mathbf{L}} \quad \bar{\mathbf{B}}] \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \end{cases} \quad (20a)$$

$$\begin{cases} \mathbf{y}_p(k+D|k) = [\mathbf{C} \quad \mathbf{0} \quad \mathbf{0}] \left((\mathbf{I} - \bar{\mathbf{M}}\bar{\mathbf{C}}) \bar{\mathbf{x}}(k|k-1) + [\bar{\mathbf{M}} \quad \mathbf{0}] \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \right) \end{cases} \quad (20b)$$

or, in a compact form,

$$\begin{cases} \bar{\mathbf{x}}(k+1|k) = \mathbf{A}_{MKP} \bar{\mathbf{x}}(k|k-1) + \mathbf{L}\mathbf{y}(k) + \bar{\mathbf{B}}\mathbf{u}(k) \end{cases} \quad (21a)$$

$$\begin{cases} \mathbf{y}_p(k+D|k) = \mathbf{C}_{MKP} \bar{\mathbf{x}}(k|k-1) + \mathbf{D}_{MKP}\mathbf{y}(k) \end{cases} \quad (21b)$$

with

$$\mathbf{A}_{MKP} = (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}),$$

$$\mathbf{C}_{MKP} = [\mathbf{C} \quad \mathbf{0} \quad \mathbf{0}] (\mathbf{I} - \bar{\mathbf{M}}\bar{\mathbf{C}}),$$

$$\mathbf{D}_{MKP} = [\mathbf{C} \quad \mathbf{0} \quad \mathbf{0}] \bar{\mathbf{M}}.$$

The KP can then be implemented using its state-space representation (21), or its matrix transfer function which is given by

$$\mathbf{Y}_p(z) = \mathcal{Z}\{\mathbf{y}_p(k+D|k)\} = \mathbf{S}_{MKP}(z)\mathbf{U}(z) + \mathbf{F}_{MKP}(z)\mathbf{Y}(z), \quad (22)$$

where

$$\mathbf{S}_{MKP}(z) = \mathbf{C}_{MKP}(z\mathbf{I} - \mathbf{A}_{MKP})^{-1}\bar{\mathbf{B}}, \quad (23)$$

$$\mathbf{F}_{MKP}(z) = \mathbf{C}_{MKP}(z\mathbf{I} - \mathbf{A}_{MKP})^{-1}\mathbf{L} + \mathbf{D}_{MKP}. \quad (24)$$

5. Kalman Predictor analysis

In [17], where the Kalman Predictor was introduced for SISO systems, it was shown that the KP implicitly defines an FSP which allows the direct use of all analysis tools of the FSP as presented in Section 2. This relation is still valid for the MKP, as stated in Theorem 1.

Theorem 1. *The Modified Kalman Predictor (MKP), defined by (21) and (22) for MIMO square processes, implicitly defines a MIMO Filtered Smith Predictor (FSP) (described by (10)) where the prediction-error filter $\mathbf{F}_{FSP}(z)$ is equal to $\mathbf{F}_{MKP}(z)$ from (24).*

Proof. See Appendix B. $\square$

This theorem makes the closed-loop analysis of the MKP in the frequency domain much more simple. Since the MIMO FSP and the MKP are equivalent, all the properties of the MIMO FSP given

in Section 2 are valid for the MKP. It is only necessary to obtain $\mathbf{F}_{MKP}(z)$, the implicit prediction-error filter of the MKP, given by (21).

Generally, for step disturbances, the $\mathbf{F}_{FSP}(z)$ static gain is chosen such that $\mathbf{F}_{FSP}(1) = \mathbf{I}$. This is necessary to guarantee that the prediction $\mathbf{y}_p(k+D|k)$ will converge to the real output $\mathbf{y}(k)$ in steady-state. From the MIMO FSP prediction Eq. (4),

$$\mathbf{Y}_p(1) = (\mathbf{I} - \mathbf{F}_{FSP}(1)\mathbf{L}(1))\mathbf{G}(1)\mathbf{U}(1) + \mathbf{F}_{FSP}(1)\mathbf{Y}$$

and since, by definition, $\mathbf{L}(1) = \mathbf{I}$, the only way for $\mathbf{Y}_p(1) = \mathbf{Y}(1)$ is if $\mathbf{F}_{FSP}(1) = \mathbf{I}$, which results in

$$\mathbf{Y}_p(1) = (\mathbf{I} - \mathbf{I})\mathbf{G}(1)\mathbf{U}(1) + \mathbf{Y}(1)$$

$$\mathbf{Y}_p(1) = \mathbf{0}\mathbf{U}(1) + \mathbf{Y}(1)$$

$$\mathbf{Y}_p(1) = \mathbf{Y}(1).$$

This property is satisfied by the MKP as shown in Corollary 1.

Corollary 1. *Given Theorem 1, the implicit prediction error filter $\mathbf{F}_{MKP}(z)$ defined by the Modified Kalman Predictor (21) for MIMO square systems has unitary static gain, that is:*

$$\mathbf{F}_{MKP}(1) = \mathbf{I}. \quad (25)$$

Proof. See Appendix C. $\square$

Another important property of the MKP has to do with its application to unstable and integrative processes. For the reasons explained in Section 2, the MIMO FSP must be tuned following some specific rules, given by (8), to avoid internal and closed-loop instability. Note that each undesirable pole will add another condition that $\mathbf{F}_{FSP}(z)$ must satisfy making the tuning procedure more involved depending of the size of the system and number of undesirable poles. The MKP, however, inherently does this independent of tuning and of the system's characteristics. This is shown in Corollary 2:

Corollary 2. *Given the detectability of system (18), Theorem 1 and the MKP defined by (21) and (22), independent of tuning, the obtained observer is stable, thus, the resulting implicit $\mathbf{S}_{MKP}(z)$ will never have any integrative or unstable open-loop poles of $\mathbf{P}(z)$.*

Proof. By Theorem 1, the MKP implicitly implements a MIMO FSP given by (10). Thus, if $\mathbf{F}_{FSP}(z) = \mathbf{F}_{MKP}(z)$, then $\mathbf{S}_{FSP}(z) = \mathbf{S}_{MKP}(z)$. In the MIMO FSP, the internal stability is obtained if $\mathbf{S}_{FSP}(z)$ is stable. However, by the definition (22) of $\mathbf{S}_{MKP}(z)$, its poles are the eigenvalues of $\mathbf{A}_{MKP} = \bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}$, and, as the observer is designed to be stable, its closed-loop poles are all inside the unitary circle. Therefore, $\mathbf{S}_{MKP}(z)$ is stable which guarantees the internal stability of the KP. $\square$

Remark 1. Note that the poles of the filter and of the $\mathbf{S}_{MKP}$ are defined with the choice of the MKP tuning parameters and, in practice, could be slow poles when robustness is required or fast poles when improved disturbance rejection is desired (as it is done in the classical MIMO FSP).

Remark 2. Another interesting point is that the KP automatically adjust the elements out of the main diagonal of the prediction error filter to obtain the desired result, whereas traditional approaches such as the MIMO FSP only use the main diagonal, which can make the tuning procedure complicated given the coupling between the variables. This happens with the MKP because the matrices $\mathbf{L}$ and $\bar{\mathbf{M}}$, which define the filter $\mathbf{F}_{MKP}(z)$ (21), generally do not have null elements. Hence, $\mathbf{F}_{MKP}(z)$, in general, will have all of its elements nonzero transfer functions, except in very specific cases.

In summary, the advantages of the proposed Modified Kalman Predictor are the following:

- With the proposed system model, the KP deals with multiple delays and directly estimates the outputs after the minimal delays without the need for integration of the model, as is common with other predictors.
- The procedure to obtain the KP is independent of the order of the transfers functions, the number of inputs/outputs, and the type of dynamics including unstable, integrative and non-minimum phase.
- Although the number of tuning parameters increases with the number of inputs/outputs, there is no need for specific guidelines for unstable, integrative or non-minimum phase processes as is the case with some of the other predictors [9, 36–39].

6. Simulation results

In this section, two examples will be shown to highlight some important characteristics of the proposed predictor. It is important to emphasize that, in this work, the problems of control and prediction are considered separately, with a focus on the latter. Hence, in the comparative examples, the same primary controllers are used and the changes will be made to the predictor structures only. This will help highlight the predictor importance in disturbance rejection and robustness. Nonetheless, ideally, these two problems should be dealt together as in [40,41]. Moreover, note that, at least in theory, it should be possible to tune different predictor structures to obtain the same closed-loop behavior for a given process. This, however, might not be trivial due to the different approaches used to obtain the predictions. This aspect is important because the main point here is not to show that a given predictor structure will have better or worse response than others, but that it is easier to implement and tune.

6.1. Example 1: Disturbance rejection tuning

In this first example, an unstable MIMO process will be used to show that the proposed predictor can cope with unstable dynamics and to show tuning guidelines regarding disturbance rejection. This process was presented in [21], and its continuous transfer matrix is given by

$$\mathbf{P}(s) = \begin{bmatrix} \frac{1.6e^{-4s}}{-2.6s+1} & \frac{0.6e^{-6s}}{2.5s+1} \\ \frac{0.7e^{-5s}}{3s+1} & \frac{1.7e^{-3s}}{-2.2s+1} \end{bmatrix}. \quad (26)$$

Discretizing with a zero-order-hold and a sampling time of $T_s = 0.2$ [s] results in

$$\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z) = \begin{bmatrix} z^{-20} & 0 \\ 0 & z^{-15} \end{bmatrix} \begin{bmatrix} \frac{-0.12793}{z-1.08} & \frac{0.04613z^{-10}}{z-0.9231} \\ \frac{0.045145z^{-10}}{z-0.9355} & \frac{-0.1618}{z-1.095} \end{bmatrix}. \quad (27)$$

To control this process, the same control structure (see Fig. 4) and tuning of [21] will be used, which was a Filtered Smith Predictor with a decoupler and PI controllers aiming at a settling time of about 12 [s] in each process variable and with low coupling. The primary controller $\mathbf{C}(z)$, reference filter $\mathbf{F}_r(z)$ and decoupler $\mathbf{D}(z)$ are given below:

$$\mathbf{C}(z) = \text{diag} \left(\frac{-2.2766(z-0.9615)}{z-1}, \frac{-2.154(z-0.9661)}{z-1} \right), \quad (28)$$

$$\mathbf{F}_r(z) = \text{diag} \left(\frac{0.25641(z-0.85)}{(z-0.9615)}, \frac{0.22599(z-0.85)}{(z-0.9661)} \right), \quad (29)$$

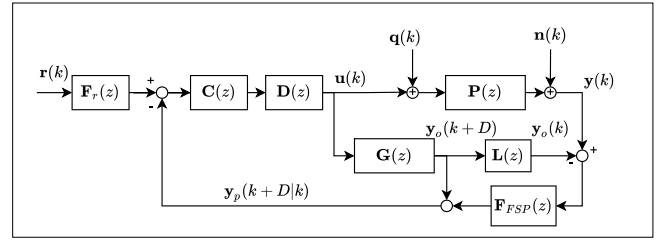


Fig. 4. Control Structure of [21] with an FSP and a decoupler $\mathbf{D}(z)$.

$$\mathbf{D}(z) = \begin{bmatrix} 1 & \frac{0.36058(z-1.08)z^{-10}}{(z-0.9231)} \\ \frac{0.27904(z-1.095)z^{-10}}{(z-0.9355)} & 1 \end{bmatrix}. \quad (30)$$

The filter of the MIMO FSP was tuned to guarantee the internal stability of the predictor, as discussed in Section 2, and to obtain an adequate trade-off between robustness and disturbance rejection:

$$\mathbf{F}_{FSP}(z) = \text{diag} \left(\frac{9.232(z-0.9892)}{(z-0.9)}, \frac{5.439(z-0.9908)}{(z-0.95)} \right) \quad (31)$$

The simulation scenario is the following: a change in the reference of 0.75 in y_1 is applied at $t = 5$ [s], a change in the reference of 0.35 in y_2 is applied at $t = 40$ [s], and input step disturbances with 0.05 of amplitude are applied at $t = 80$ [s], and $t = 120$ [s] in inputs u_1 and u_2 , respectively. Since, in this example, only the nominal case will be dealt with, the reference tracking is always the same, regardless of the predictor and its tuning, as explained earlier. For clarity, the simulation results are divided into two figures. The reference tracking portion of the simulation scenario is shown in Fig. 5 and the disturbance rejection responses are shown in Fig. 6 for six different cases chosen to highlight and explain the tuning procedure of the Modified Kalman Predictor. For the reference tracking, only one case is shown because the predictors do not change the nominal reference-tracking response. Notice that the response has a settling time of approximately 12 [s] after the dead time. Each case that will be presented has a different predictor or a different tuning:

- Case FSP: uses the MIMO Filtered Smith Predictor of [21] described earlier.
- Case 0: the base case for comparison. MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.
- Case Y1: worse disturbance rejection for y_1 . MKP with $\mathcal{R} = \text{diag}(10, 1)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.
- Case Y2: worse disturbance rejection for y_2 . MKP with $\mathcal{R} = \text{diag}(1, 10)$, $\mathcal{Q} = \text{diag}(0.05, 0.05)$.
- Case Q1: better disturbance rejection for q_1 . MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(5, 0.05)$.
- Case Q2: better disturbance rejection for q_2 . MKP with $\mathcal{R} = \text{diag}(1, 1)$, $\mathcal{Q} = \text{diag}(0.05, 5)$.

Case FSP and Case 0 will be used for comparison, with Case 0 being the base case for the MKP. In cases Y1 and Y2 only the weights $\mathcal{R}$ associated with the outputs change. In Case Y1, the weight associated with y_1 is increased a tenfold, indicating a higher variance with this variable, making the predictor less confident on this measurement. In Case Y2, the same is done for y_2 . The results can be seen in Fig. 6(a).

To make the comparison easier, the Integral Squared Error (ISE) was computed, which has the following definition for a generic signal x :

$$\delta(t_i, t_f, x) = \int_{t_i}^{t_f} (x_{\text{ref}}(t) - x(t))^2 dt, \quad (32)$$

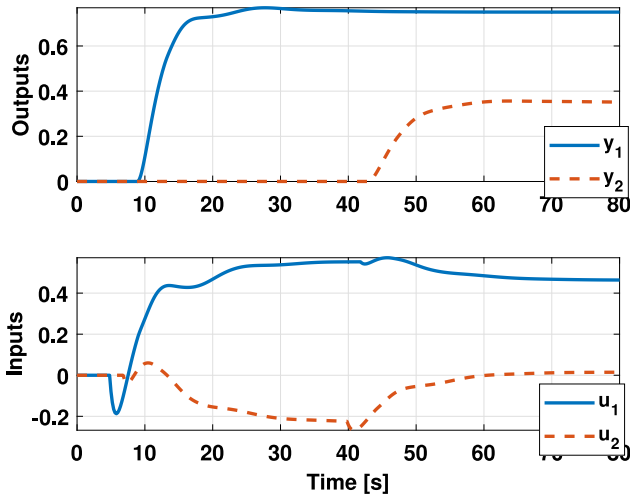


Fig. 5. Example 1 - nominal reference tracking for all cases are identical.

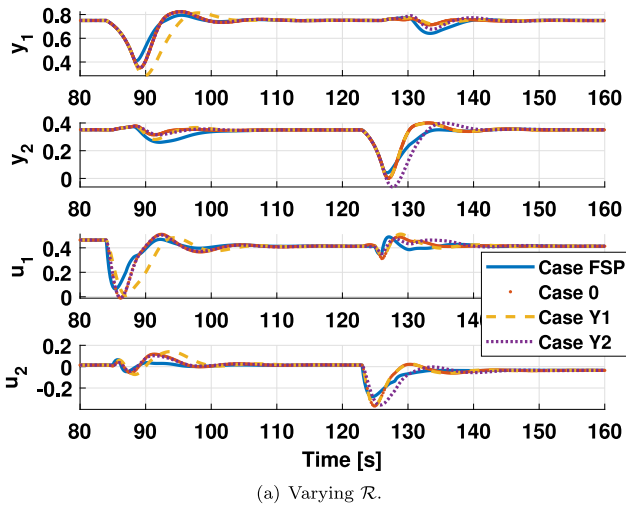
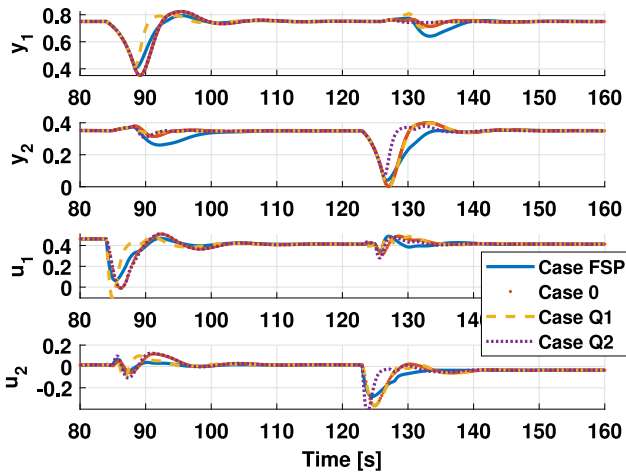
(a) Varying $\mathcal{R}$.(b) Varying $\mathcal{Q}$.

Fig. 6. Example 1 - disturbance rejection for different predictors and tuning of the MKP.

Table 1

Normalized Integral Squared Errors (ISE) for Example 1.

Case	δ_{y_1}	δ_{y_2}	δ_{q_1}	δ_{q_2}
Case FSP	0.7470	1.0935	0.7405	1.1041
Case 0	1.0000	1.0000	1.0000	1.0000
Case Y1	1.8425	1.0217	1.8589	0.9952
Case Y2	1.0273	1.9317	0.9979	1.9781
Case Q1	0.4603	1.0124	0.4428	1.0400
Case Q2	1.0014	0.4197	1.0073	0.4096

where x_{ref} is the reference for x while t_i and t_f indicate the initial and final time of evaluation, respectively.¹ Considering the ISE index characteristics, a better performance is associated with smaller values of δ , since it indicates smaller errors. The quantitative values based in (32), normalized in relation to Case 0, are shown in Table 1. Thus, δ values smaller than 1 indicate a better performance than Case 0, and the opposite is true for values greater than 1. In the table, $\delta_{y_1} = \delta(80, 160, y_1)$ and $\delta_{y_2} = \delta(80, 160, y_2)$ reflect the disturbance rejection of each output while $\delta_{q_1} = \delta(80, 120, y_1) + \delta(80, 120, y_2)$ and $\delta_{q_2} = \delta(120, 160, y_1) + \delta(120, 160, y_2)$ reflect the disturbance rejection in all outputs for a particular disturbance.

From Fig. 6(a), note that the disturbance rejection for Case Y1 is slower for y_1 in comparison with the base case, whereas for y_2 the response is almost the same. This is also reflected in δ_{y_1} in Table 1, which increases 34% in relation to Case 0, while δ_{y_2} increases only 8%. The converse is true for Case Y2. Given the coupled nature of the process, a completely decoupled tuning is very hard for any predictor but, with the KP, the coupling effects in the tuning are minimized without any special consideration.

This behavior is reflected in the Bode Diagram of the implicit prediction error filter of the KP which is shown in Fig. 7(a), and where e_{yi} is the i th input of the filter and the i th prediction error, and e_{fyi} is the filtered prediction error and i th output of the filter. With Case Y1, the part of the filter associated with y_1 ($e_{y1} - e_{fy1}$) becomes less high pass, which indicates a slower dynamics in relation to Case 0, while the part associated with y_2 ($e_{y2} - e_{fy2}$) almost does not change. The converse is true for Case Y2. The Bode diagram also confirms Remark 2, which states that the KP automatically adjust the elements of the filter out of the main diagonal to obtain the desired result. It is also important to emphasize that, in the Bode diagram, plots with higher peaks and high magnitude in high frequencies imply faster responses and lower robustness, and vice-versa.

In cases Q1 and Q2 the weights associated with the disturbances $\mathcal{Q}$ are varied. In Case Q1, the weight associated with q_1 is increased a hundredfold to 5, indicating a higher variance with this variable, making the predictor react more aggressively in the estimation of q_1 . In Case Q2, the same is done but with q_2 . The results can be seen in Fig. 6(b). Note that rejection of the disturbance in u_1 ($t = 80$ [s]) is faster in both output variables, whereas the rejection of the disturbance in u_2 ($t = 120$ [s]) remains practically the same for both outputs. This is also reflected in δ_{q_1} in Table 1 which decreases 35% in relation to Case 0, whereas δ_{q_2} actually increases a little (6.5%). The converse is true for Case Q2.

As in cases Y1 and Y2, this behavior is reflected in the Bode Diagram of the implicit prediction error filter of the KP (Fig. 7(b)). With Case Q1, the part of the filter associated with y_1 ($e_{y1} - e_{fy1}$) becomes more high-pass, which indicates a faster dynamics in relation to Case 0, while the part associated with y_2 ($e_{y2} - e_{fy2}$) almost does not change. The converse is true for Case Q2.

¹ Other performance indices could be used, i.e., Integral Absolute Error (IAE) or Integral Time Absolute Error (ITAE), but, in our example, they all resulted in the same conclusions and were omitted.

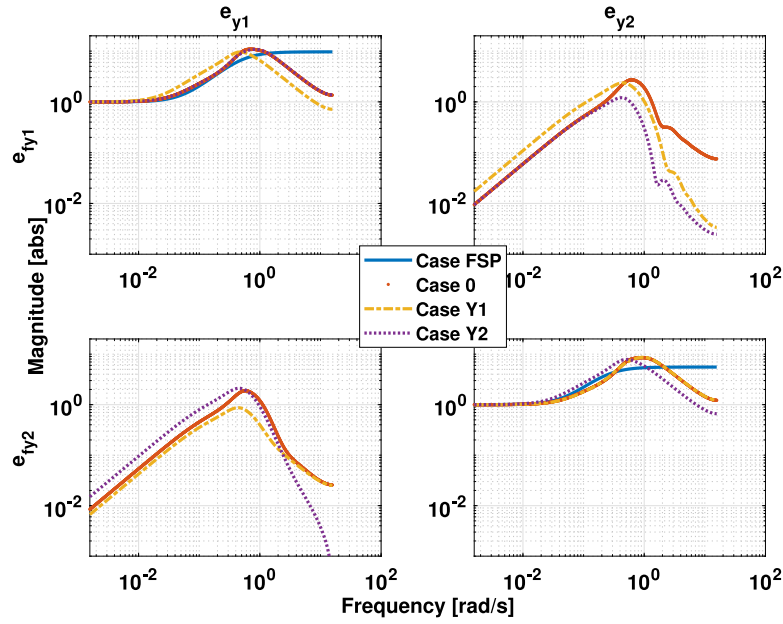
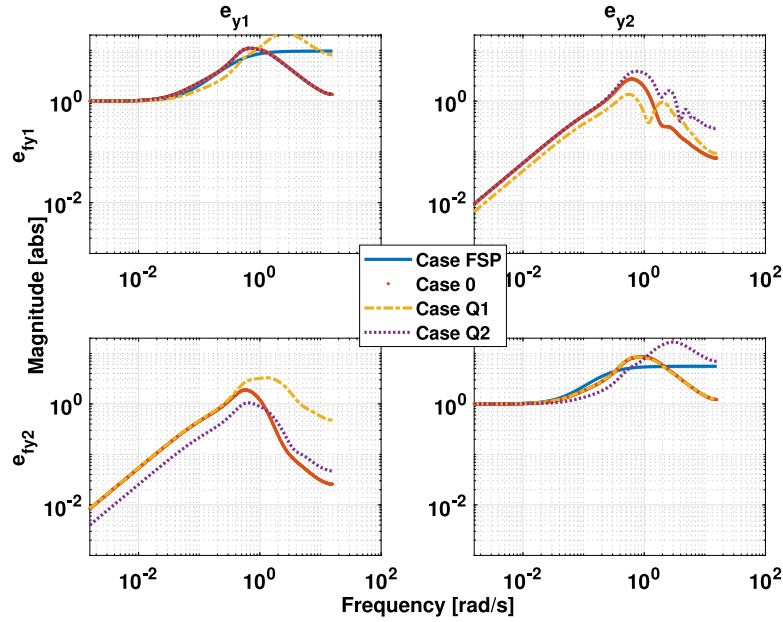
(a) Varying $\mathcal{R}$.(b) Varying $\mathcal{Q}$.

Fig. 7. Example 1 - Bode Diagram of the different prediction error filters.

From these results, note that the KP is capable of handling even systems with unstable dynamics without any change to its formulation, which is an important property. Also, its tuning parameters allows the adjustment of the closed-loop response considering each output and input disturbance individually, giving a good number of degrees of freedom to adjust the performance of the closed-loop system. For example, to improve the disturbance rejection for disturbance i , the q_i element of matrix $\mathcal{Q}$ must be bigger than the rest of q_j , $\forall j \neq i$. If we intend to worsen the disturbance rejection for output y_i , the r_i element of matrix $\mathcal{R}$ must be bigger than the rest of r_j , $\forall j \neq i$. It is important to note that this tuning procedure does not guarantee a perfect decoupling effect, and it is expected that, although all variables are affected by the tuning choices, the selected ones will be the

most affected. Finally, the KP tuning does not depend on the order of the system or its dynamics.

Also, from this example, one can draw some conclusion that are useful to tune the Kalman Predictor:

- Decreasing the weight associated with a given output variable y_i in $\mathcal{R}$ improves the disturbance rejection associated with this variable.
- Increasing the weight associated with a given disturbance q_i in $\mathcal{Q}$ improves the disturbance rejection associated with the i th input.

These are heuristics guidelines but, coupled with a frequency domain analysis of the predictor, it is possible to tune the predictor adequately to obtain the desired response. Also, it is important to emphasize that a better disturbance rejection usually results

in a worse robustness for the closed-loop system, thus, a careful analysis is important in each specific case, as will be shown in the next example.

6.2. Example 2: Robustness tuning

The example presented in this section is the Shell's Heavy Oil Fractionator, a process widely used in the literature to test control structures [1,42,43]. This problem will be used to show how the Kalman Predictor can be used to improve the robustness of a control structure.

The controlled variables of this process are: the top (y_1) and side (y_2) product compositions measured by analyzers, and the bottom temperature (y_3). The manipulated variables are: the top draw rate (u_1), the side draw rate (u_2) and the bottom reflux duty (u_3). The upper reflux duty (q_1) is considered an unmeasurable disturbance. Near the operating point, the dynamics of the process can be described by the following model:

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \\ Y_3(s) \end{bmatrix} = \begin{bmatrix} \frac{4.05e^{-28s}}{50s+1} & \frac{1.77e^{-28s}}{60s+1} & \frac{5.88e^{-28s}}{50s+1} \\ \frac{5.39e^{-18s}}{50s+1} & \frac{5.72e^{-14s}}{60s+1} & \frac{6.9e^{-16s}}{40s+1} \\ \frac{4.38e^{-20s}}{33s+1} & \frac{4.42e^{-22s}}{44s+1} & \frac{7.2}{19s+1} \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \\ U_3(s) \end{bmatrix} + \begin{bmatrix} \frac{1.44e^{-27s}}{1+40s} \\ \frac{1.83e^{-15s}}{1+20s} \\ \frac{1.26}{1+32s} \end{bmatrix} Q_1(s). \quad (33)$$

The discrete model was obtained with discretizing with a zero-order-hold and a sampling time of $T_s = 2$ [min].

The simulation scenario is the following: at $t = 0$ [min], the references of y_1 , y_2 and y_3 are changed to 0.5, 0.3 and 0.1, respectively; at $t = 100$ [min] the y_1 reference is changed to 0.4; the disturbance q_1 changes its value from 0 to 0.02 at $t = 200$ [min].

For this process, the primary controller will be a GPC [44] with the following tuning parameters: weighting factors for error tracking $\delta_1 = \delta_2 = \delta_3 = 1$; weighting factor for control increments $\lambda_1 = \lambda_2 = 0.3$, $\lambda_3 = 0.5$; control horizons $N_{u1} = N_{u2} = N_{u3} = 30$; prediction horizons $N_1 = N_2 = N_3 = 60$ after each minimal output delay, which gives a settling time of about 13 [min] for each output after the minimum dead-time. Three cases will be considered to highlight the robustness properties of the MKP while using the same primary controller based on the GPC:

- Case GPC: the predictor here is the implicit one defined by the GPC algorithm when the nominal plant $P(z)$ is used [1].
- Case MKP1: the implicit GPC predictor is substituted by the MKP, which is tuned to achieve a similar result to the previous case.² $Q = \text{diag}(0.1, 3, 50)$, $R = \text{diag}(0.1, 0.1, 0.5)$.
- Case MKP2: the same as Case MKP1, but tuned for robustness. $Q = I$, $R = 13\,000I$.

In Fig. 8, the nominal case is presented. Note that the reference tracking for all cases is the same, as was expected, and that disturbance rejection of Case GPC and Case KP1 are similar. Case KP2 has a slower disturbance rejection, as expected from the tuning. To show this quantitatively, the Integral Time Absolute Error (ITAE) was calculated for each case, and then normalized in relation to Case GPC. The ITAE was used here instead of the ISE

Table 2

Normalized Integral Time Absolute Errors (ITAE) for Example 2.

Case	ITAE
Case GPC	1.0000
Case MKP1	0.9961
Case MKP2	2.7217

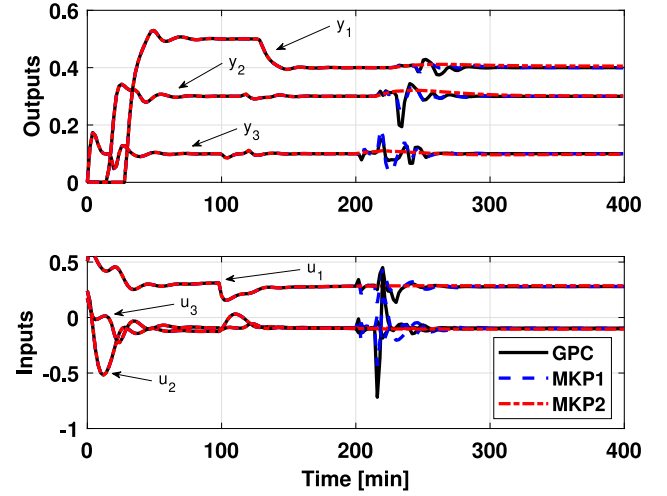


Fig. 8. Example 2 - reference tracking and disturbance rejection in the nominal case.

because it represented more accurately the slow performance of Case MKP2. The ITAE for a vector $\mathbf{x}$ is calculated as:

$$\text{ITAE}(t_i, t_f, \mathbf{x}) = \int_{t_i}^{t_f} t \|\mathbf{x}_{\text{ref}}(t) - \mathbf{x}(t)\| dt,$$

where $t_i = 200$, $t_f = 400$, so that the performance index is computed only for the disturbance rejection, since the reference tracking is the same. The ITAE values are shown in Table 2, where smaller values indicate a better disturbance rejection.

An SVD analysis of the frequency domain characteristics of the prediction error filters was made and is presented in Fig. 9. The filter from Case GPC has higher gain at high frequencies if compared to Case KP1 which presents similar disturbance rejection. In other words, Case KP1 has better robustness with similar performance. Case KP2 has low-pass characteristics which improves robustness at the cost of disturbance rejection performance.

To verify the robust performance, consider that there is a model mismatch in the delays of (33). The delay is 30 in element (1, 2) of $\mathbf{P}(s)$ instead of 28 used in the model of the predictor and 16 instead of 18 in element (2, 1). In both cases this translates in a one sample delay mismatch in the discretized model. Considering an additive modeling of the model mismatch, the control scheme must satisfy the robustness condition (11) which is presented in Fig. 10 for all cases, given that $\mathbf{W}_a(z) = \mathbf{W}_2(z)\Delta(z)\mathbf{W}_1$, $\mathbf{W}_2 = I$ and $\mathbf{W}_1 = \mathbf{P}(z) - \mathbf{P}_n(z)$, where $\mathbf{P}_n(z)$ is the nominal model. From Fig. 10, it is clear that Case GPC and Case KP1 do not satisfy the robustness condition, hence, closed-loop stability for this set of uncertainties is not guaranteed. Case KP2 does satisfy the condition, meaning that robust stability is achieved for this set of uncertainties. The simulation results of the uncertain case (Fig. 11) corroborate this analysis, only Case KP2 is stable.

Hence, the results from this example have confirmed that the predictor structure has an important role in the closed-loop robustness. In the MPC context, this was originally addressed in [1], where it was proven that the implicit predictor of some MPC algorithms provide very little robustness to model uncertainties. But this behavior can be improved by substituting the

² The GPC is tuned using the model of the process without the minimal delays.

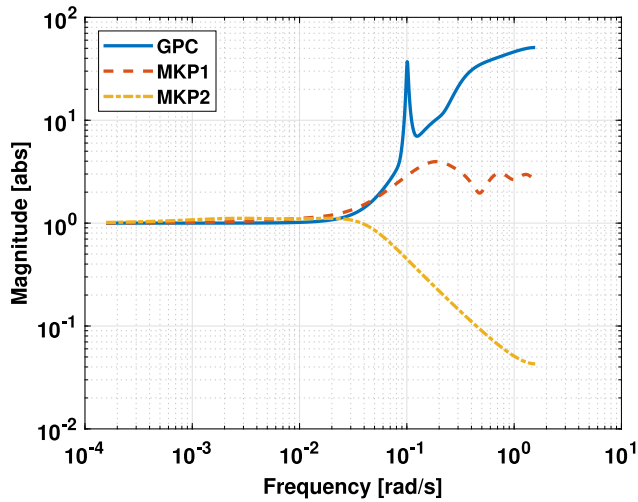


Fig. 9. Example 2 - prediction error filters SVD ($\hat{\sigma}(\cdot)$) analysis.

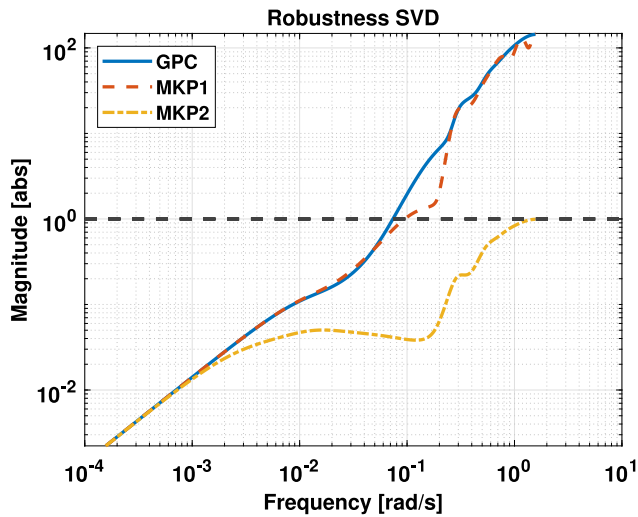


Fig. 10. Example 2 - robustness analysis using (11). Gray dashed line indicates robustness condition limit.

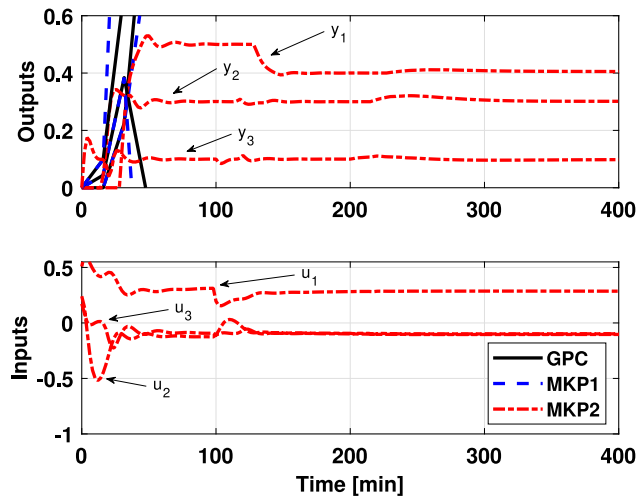


Fig. 11. Example 2 - reference tracking and disturbance rejection in the uncertain case. GPC and MKP1 cases are unstable and the simulation of these two cases was stopped at $t = 50$ min approximately and only the process outputs are shown.

implicit predictor by an SP or FSP based strategy, just as was made here with the MKP. With the proposed MIMO KP, it becomes easier to adjust the system performance without changing the primary controller and without using more complex computations to define the internal stability properties of the predictor, as is the case with other predictors such as the MIMO FSP. Thus, for practitioners, the MKP seems to be an interesting approach for predictor based controllers in MIMO delayed systems. Finally, although the use of a separate tuning strategy for the predictor and primary controller is useful in practice, it is important to emphasize that, in some situations better results can be achieved when both primary controller and predictor structure are tuned together. This can be a topic of future research in the context of the MKP.

7. Conclusions

The Modified Kalman Predictor (MKP) presented in this work adapts the predictor shown in [17] to multiple-delay-MIMO square systems of any order and dynamics. The MKP has intuitive tuning guidelines and it allows the use of traditional frequency domain analysis tools to verify its closed-loop properties, such as disturbance rejection and robustness, as shown through two examples. Its formulation provides a straight-forward approach to obtain the prediction of the system that does not require integration of the model. Moreover, as its structure is general and does not require complex conditions, its implementation and tuning is easy, allowing to achieve a compromise between robustness and disturbance rejection.

It is important to emphasize that other predictor structures can also be tuned to offer improved disturbance rejection or improved robustness similarly to the MKP, but it is argued that the tuning procedure is not as easy, and many of these structures do not cope with multiple delays.

One of its limitations is that it is dependent of the minimal output delays of the system. If they are small in relation to the remaining delays, the advantages of the MKP are reduced since the order of the fast model $G(z)$ will not be reduced significantly in relation to the full process model $P(z)$. Another issue is that, in its current form, it can only be applied to square systems and only step disturbances are considered in the formulation. Furthermore, for bigger system, there might be some numerical issues in the computation of the Kalman Filter. These limitations will be the subject of future research.

CRedit authorship contribution statement

Daniel Martins Lima: Conceptualization, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Supervision, Funding acquisition. **Bruno Martins Lima:** Conceptualization, Formal analysis, Resources, Writing – original draft. **Julio Elias Normey-Rico:** Conceptualization, Formal analysis, Resources, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Daniel Martins Lima reports financial support was provided by National Council for Scientific and Technological Development. Julio Elias Normey-Rico reports financial support was provided by National Council for Scientific and Technological Development.

Data availability

No data was used for the research described in the article.

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Appendix A. State-space Filtered Smith Predictor

Analyzing the block diagram of the Filtered Smith Predictor shown in Fig. 2, and using the process model Eqs. (12) and (13), it is easy to show that the prediction is given by

$$\mathbf{y}_p(k+D|k) = \mathbf{y}_o(k+D) + \mathbf{y}_f(k) \quad (\text{A.1})$$

where $\mathbf{y}_o(k+D)$ is the output without the minimal delays obtained from the nominal model and $\mathbf{y}_f(k)$ is the output of the prediction error filter $\mathbf{F}_{FSP}(z)$, whose dynamics can be represented in state space form as

$$\begin{cases} \mathbf{x}_f(k+1) = \mathbf{A}_f \mathbf{x}_f(k) + \mathbf{B}_f (\mathbf{y}(k) - \mathbf{y}_o(k)) \\ \mathbf{y}_f(k) = \mathbf{C}_f \mathbf{x}_f(k) + \mathbf{D}_f (\mathbf{y}(k) - \mathbf{y}_o(k)) \end{cases} \quad (\text{A.2})$$

where $\mathbf{y}$ is the measured output, $\mathbf{y}_o$ is the output of the nominal model $\mathbf{P}(z) = \mathbf{L}(z)\mathbf{G}(z)$, $\mathbf{x}_f$ is the filter state, and matrices $\mathbf{A}_f$, $\mathbf{B}_f$, $\mathbf{C}_f$ and $\mathbf{D}_f$ represent the dynamics of $\mathbf{F}_{FSP}(z)$. The complete dynamics of the FSP is obtained by combining the previous equation with the process model Eqs. (12) and (13), which results in

$$\begin{cases} \begin{bmatrix} \mathbf{x}(k+1) \\ \mathbf{x}_f(k+1) \\ \mathbf{x}_f(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_f \mathbf{C} & \mathbf{A}_f & \mathbf{0} \\ -\mathbf{B}_f \mathbf{D}_f \mathbf{C} & -\mathbf{B}_f \mathbf{C}_f & \mathbf{A}_f \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_f(k) \\ \mathbf{x}_f(k) \end{bmatrix} \\ \quad + \begin{bmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{B}_f & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \\ \mathbf{y}_p(k+D|k) = \begin{bmatrix} (\mathbf{C} - \mathbf{D}_f \mathbf{D}_f \mathbf{C}) & -\mathbf{D}_f \mathbf{C}_f & \mathbf{C}_f \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{x}_f(k) \\ \mathbf{x}_f(k) \end{bmatrix} \\ \quad + \begin{bmatrix} \mathbf{D}_f & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{y}(k) \\ \mathbf{u}(k) \end{bmatrix} \end{cases} \quad (\text{A.3})$$

Applying the Z-transform to this state-space representation and after some matrix manipulations, the following is obtained:

$$\mathbf{Y}_p(z) = \mathbf{S}_{FSP}(z)\mathbf{U}(z) + \mathbf{F}_{FSP}\mathbf{Y}(z), \quad (\text{A.4})$$

where

$$\mathbf{F}_{FSP}(z) = \mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f, \quad (\text{A.5})$$

$$\begin{aligned} \mathbf{S}_{FSP}(z) &= \{ \mathbf{I} - (\mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f) (\mathbf{C}_l (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l + \mathbf{D}_l) \} \\ &\quad \times \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}, \\ &= (\mathbf{I} - \mathbf{F}_{FSP}(z)\mathbf{L}(z)) \mathbf{G}(z). \end{aligned} \quad (\text{A.6})$$

Appendix B. Proof of Theorem 1

From (22), the transfer functions associated with the MKP are given by:

$$\mathbf{F}_{MKP}(z) = \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} ((\mathbf{I} - \mathbf{M}\bar{\mathbf{C}})(z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \mathbf{L} + \mathbf{M}) \quad (\text{B.1})$$

$$\mathbf{S}_{MKP}(z) = \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}})(z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \bar{\mathbf{B}} \quad (\text{B.2})$$

Comparing (B.1) and (A.5), it is easy to see that they are equal if $\mathbf{F}_{FSP}(z)$ is defined by:

$$\begin{aligned} \mathbf{D}_f &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{M}, & \mathbf{C}_f &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}) \\ \mathbf{B}_f &= \mathbf{L}, & \mathbf{A}_f &= \bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}} \end{aligned} \quad (\text{B.3})$$

To complete the proof, it is necessary to show that if $\mathbf{F}_{FSP}(z) = \mathbf{F}_{MKP}(z)$ then $\mathbf{S}_{PSF}(z) = \mathbf{S}_{MKP}(z)$. For this step, consider the Woodbury matrix identity

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - (\mathbf{A} + \mathbf{BCD})^{-1} \mathbf{BCD} \mathbf{A}^{-1}, \quad (\text{B.4})$$

which requires an invertible $\mathbf{A}$ and $\mathbf{B}$, and also $\mathbf{C}$ and $\mathbf{D}$ of appropriate dimensions. This can be applied to $(z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1}$, resulting in

$$(z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} = (z\mathbf{I} - \bar{\mathbf{A}})^{-1} - (z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \bar{\mathbf{L}}\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1}. \quad (\text{B.5})$$

Applying this to (B.2), results in

$$\begin{aligned} \mathbf{S}_{MKP}(z) &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}) ((z\mathbf{I} - \bar{\mathbf{A}})^{-1} - (z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \\ &\quad \times \bar{\mathbf{L}}\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1}) \bar{\mathbf{B}} \\ &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}}) ((z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} - (z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \\ &\quad \times \bar{\mathbf{L}}\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}}) \\ &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} - \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{M}\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \\ &\quad - \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}})(z\mathbf{I} - (\bar{\mathbf{A}} - \bar{\mathbf{L}}\bar{\mathbf{C}}))^{-1} \bar{\mathbf{L}}\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \end{aligned} \quad (\text{B.6})$$

Using the equivalency given by (B.7), shown Box 1, we have:

$$(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} = \begin{bmatrix} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \quad (\text{B.8})$$

$$\bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} = (\mathbf{C}_l (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l + \mathbf{D}_l) \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \quad (\text{B.9})$$

Applying these last three equations to (B.6) and using the definitions in (B.3) results in

$$\begin{aligned} \mathbf{S}_{MKP}(z) &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} - \mathbf{D}_f \bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \\ &\quad - \mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f \bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \\ &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \\ &\quad - (\mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f) \bar{\mathbf{C}}(z\mathbf{I} - \bar{\mathbf{A}})^{-1} \bar{\mathbf{B}} \\ &= \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \\ &\quad - (\mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f) (\mathbf{C}_l (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l + \mathbf{D}_l) \\ &\quad \times \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ &= \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ &\quad - (\mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f) (\mathbf{C}_l (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l + \mathbf{D}_l) \\ &\quad \times \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ &= \{ \mathbf{I} - (\mathbf{C}_f (z\mathbf{I} - \mathbf{A}_f)^{-1} \mathbf{B}_f + \mathbf{D}_f) (\mathbf{C}_l (z\mathbf{I} - \mathbf{A}_l)^{-1} \mathbf{B}_l + \mathbf{D}_l) \} \\ &\quad \times \mathbf{C} (z\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \\ &= (\mathbf{I} - \mathbf{F}_{MKP}(z)\mathbf{L}(z)) \mathbf{G}(z) \end{aligned}$$

which completes the proof since this last definition is equal to $\mathbf{S}_{FSP}(z)$ in (A.6). $\square$

Appendix C. Proof of Corollary 1

From the definition of $\mathbf{S}_{FSP}(z)$ in (A.4), if $\mathbf{F}_{FSP}(1) = \mathbf{I}$, then $\mathbf{S}_{FSP}(1) = \mathbf{0}$, as explained in Section 5. Since the MKP implicitly defines an FSP as stated in Theorem 1, then, if $\mathbf{S}_{MKP}(1) = \mathbf{0}$ is true, it implies that $\mathbf{F}_{MKP}(1) = \mathbf{I}$. From (22),

$$\mathbf{S}_{MKP}(z) = \begin{bmatrix} \mathbf{C} & \mathbf{0} & \mathbf{0} \end{bmatrix} (\mathbf{I} - \mathbf{M}\bar{\mathbf{C}})(z\mathbf{I} - \bar{\mathbf{A}} + \bar{\mathbf{L}}\bar{\mathbf{C}})^{-1} \bar{\mathbf{B}}. \quad (\text{C.1})$$

Partitioning $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, $\bar{\mathbf{C}}$, $\mathbf{M}$ and $\mathbf{L}$ to make the notation easier, it follows that

$$\bar{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & \mathbf{0} & \mathbf{B} \\ \mathbf{B}_l \mathbf{C} & \mathbf{A}_l & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_x & \mathbf{B}_x \\ \mathbf{0} & \mathbf{I} \end{bmatrix},$$

$$(z\mathbf{I} - \bar{\mathbf{A}})^{-1} = \begin{bmatrix} (z\mathbf{I} - \mathbf{A})^{-1} & \mathbf{0} & (z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{C}_\phi(z\mathbf{I} - \mathbf{A}_\phi)^{-1} \\ (z\mathbf{I} - \mathbf{A}_1)^{-1}\mathbf{B}_1\mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1} & (z\mathbf{I} - \mathbf{A}_1)^{-1} & (z\mathbf{I} - \mathbf{A}_1)^{-1}\mathbf{B}_1\mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{C}_\phi(z\mathbf{I} - \mathbf{A}_\phi)^{-1} \\ \mathbf{0} & \mathbf{0} & (z\mathbf{I} - \mathbf{A}_\phi)^{-1} \end{bmatrix} \quad (\text{B.7})$$

Box I.

$$\begin{aligned} (z\mathbf{I} - \bar{\mathbf{A}} + \mathbf{L}\bar{\mathbf{C}})^{-1} &= \begin{bmatrix} \mathbf{A}_1^{-1} \left(\mathbf{I} + \mathbf{B}_1 (\mathbf{D}_1 - \mathbf{C}_1\mathbf{A}_1^{-1}\mathbf{B}_1)^{-1} \mathbf{C}_1\mathbf{A}_1^{-1} \right) & -\mathbf{A}_1^{-1}\mathbf{B}_1 (\mathbf{D}_1 - \mathbf{C}_1\mathbf{A}_1^{-1}\mathbf{B}_1)^{-1} \\ -(\mathbf{D}_1 - \mathbf{C}_1\mathbf{A}_1^{-1}\mathbf{B}_1) \mathbf{C}_1\mathbf{A}_1^{-1} & (\mathbf{D}_1 - \mathbf{C}_1\mathbf{A}_1^{-1}\mathbf{B}_1)^{-1} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{A}_2 & \mathbf{B}_2 \\ \mathbf{C}_2 & \mathbf{D}_2 \end{bmatrix}. \end{aligned} \quad (\text{C.3})$$

Box II.

$$\bar{\mathbf{B}} = \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{B}_x \\ \mathbf{0} \end{bmatrix}, \quad \bar{\mathbf{C}} = [\mathbf{D}_1\mathbf{C} \quad \mathbf{C}_1 \quad \mathbf{0}] = [\mathbf{C}_x \quad \mathbf{0}], \quad (\text{C.2})$$

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_x \\ \mathbf{L}_\phi \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} \mathbf{M}_x \\ \mathbf{M}_\phi \end{bmatrix}, \quad [\mathbf{C} \quad \mathbf{0} \quad \mathbf{0}] = [\mathbf{C}_{x2} \quad \mathbf{0}],$$

then,

$$\begin{aligned} (z\mathbf{I} - \bar{\mathbf{A}} + \mathbf{L}\bar{\mathbf{C}}) &= z\mathbf{I} - \begin{bmatrix} \mathbf{A}_x & \mathbf{B}_x \\ \mathbf{0} & \mathbf{I} \end{bmatrix} + \begin{bmatrix} \mathbf{L}_x \\ \mathbf{L}_\phi \end{bmatrix} [\mathbf{C}_x \quad \mathbf{0}] \\ &= \begin{bmatrix} z\mathbf{I} - \mathbf{A}_x & -\mathbf{B}_x \\ \mathbf{0} & z\mathbf{I} - \mathbf{I} \end{bmatrix} + \begin{bmatrix} \mathbf{L}_x\mathbf{C}_x & \mathbf{0} \\ \mathbf{L}_\phi\mathbf{C}_x & \mathbf{0} \end{bmatrix} \\ &= \begin{bmatrix} z\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x & -\mathbf{B}_x \\ \mathbf{L}_\phi\mathbf{C}_x & z\mathbf{I} - \mathbf{I} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{B}_1 \\ \mathbf{C}_1 & \mathbf{D}_1 \end{bmatrix}. \end{aligned}$$

Using some matrix properties of inverse block matrices, we have (C.3), shown in Box II. Substituting (C.3) and (C.2) in (C.1),

$$\begin{aligned} \mathbf{S}_{MKP}(z) &= [\mathbf{C}_{x2} \quad \mathbf{0}] \left(\mathbf{I} - \begin{bmatrix} \mathbf{M}_x \\ \mathbf{M}_\phi \end{bmatrix} [\mathbf{C}_x \quad \mathbf{0}] \right) \begin{bmatrix} \mathbf{A}_2 & \mathbf{B}_2 \\ \mathbf{C}_2 & \mathbf{D}_2 \end{bmatrix} \begin{bmatrix} \mathbf{B}_x \\ \mathbf{0} \end{bmatrix} \\ &= [\mathbf{C}_{x2} \quad \mathbf{0}] \begin{bmatrix} \mathbf{I} - \mathbf{M}_x\mathbf{C}_x & \mathbf{0} \\ -\mathbf{M}_\phi\mathbf{C}_x & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{A}_2\mathbf{B}_x \\ \mathbf{C}_2\mathbf{B}_x \end{bmatrix} \\ &= [\mathbf{C}_{x2}(\mathbf{I} - \mathbf{M}_x\mathbf{C}_x) \quad \mathbf{0}] \begin{bmatrix} \mathbf{A}_2\mathbf{B}_x \\ \mathbf{C}_2\mathbf{B}_x \end{bmatrix} \\ &= \mathbf{C}_{x2}(\mathbf{I} - \mathbf{M}_x\mathbf{C}_x)\mathbf{A}_2\mathbf{B}_x \end{aligned}$$

So, it is only necessary to compute the term $\mathbf{A}_2$ from (C.3), which is given by

$$\begin{aligned} \mathbf{A}_2 &= \mathbf{A}_1^{-1} \left(\mathbf{I} + \mathbf{B}_1 (\mathbf{D}_1 - \mathbf{C}_1\mathbf{A}_1^{-1}\mathbf{B}_1)^{-1} \mathbf{C}_1\mathbf{A}_1^{-1} \right) \\ &= (z\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1} \\ &\quad \times \left(\mathbf{I} - \mathbf{B}_x ((z\mathbf{I} - \mathbf{I}) + \mathbf{L}_\phi\mathbf{C}_x(z\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1}\mathbf{B}_x)^{-1} \right. \\ &\quad \times \mathbf{L}_\phi\mathbf{C}_x(z\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1} \left. \right). \end{aligned}$$

Then, $\mathbf{S}_{MKP}(1)$ is given by

$$\begin{aligned} \mathbf{S}_{MKP}(1) &= \mathbf{C}_{x2}(\mathbf{I} - \mathbf{M}_x\mathbf{C}_x)(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1} \\ &\quad \times \left(\mathbf{I} - \mathbf{B}_x ((\mathbf{I} - \mathbf{I}) + \mathbf{L}_\phi\mathbf{C}_x(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1}\mathbf{B}_x)^{-1} \right. \\ &\quad \times \mathbf{L}_\phi\mathbf{C}_x(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1} \left. \right) \mathbf{B}_x \\ \mathbf{S}_{MKP}(1) &= \mathbf{C}_{x2}(\mathbf{I} - \mathbf{M}_x\mathbf{C}_x)(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1} \\ &\quad \times \left(\mathbf{B}_x - \mathbf{B}_x (\mathbf{L}_\phi\mathbf{C}_x(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1}\mathbf{B}_x)^{-1} \right. \\ &\quad \times \mathbf{L}_\phi\mathbf{C}_x(\mathbf{I} - \bar{\mathbf{A}} + \mathbf{L}\bar{\mathbf{C}})^{-1}\mathbf{B}_x \left. \right), \end{aligned}$$

$$\mathbf{S}_{MKP}(1) = \mathbf{C}_{x2}(\mathbf{I} - \mathbf{M}_x\mathbf{C}_x)(\mathbf{I} - \mathbf{A}_x + \mathbf{L}_x\mathbf{C}_x)^{-1}(\mathbf{B}_x - \mathbf{B}_x) = \mathbf{0},$$

which completes the proof. $\square$

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